Quantification: 535_z27c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	16
interior lacuna count	16
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	3620.69	2586.5	2993.42	px^2
major_axis_length_px	lacuna shape	97.0969	98.0661	41.6941	px
minor_axis_length_px	lacuna shape	49.6302	45.6213	21.8403	px
aspect_ratio	lacuna shape	2.0421	2.1684	0.6922	unitless
eccentricity	lacuna shape	0.8193	0.8873	0.1342	unitless
solidity	lacuna shape	0.804	0.8071	0.1018	unitless
roots_count	headline	7.875	7	4.2876	count
ring_length_r30_px	headline	262.562	228	118.073	px
ring_length_r60_px	reported	580.438	488.5	265.982	px
owned_length_px	ownership-dependent	735.477	443.44	630.818	px
edge_count	ownership-dependent	32.1875	17	28.1027	count
mean_edge_length_px	ownership-dependent	24.955	24.2772	5.369	px
perimeter_px	normalised	315.423	268.01	156.463	px
ring_area_r30_px2	normalised	10053.3	9476	4228.66	px^2
ring_area_r60_px2	normalised	21570.2	20265	8543.91	px^2
in_frame_fraction_r30	normalised	0.9917	1	0.0331	unitless
in_frame_fraction_r60	normalised	0.9851	1	0.0597	unitless
ring_density_r30	normalised	0.026338	0.0264178	0.00599012	px^-1
ring_density_r60	normalised	0.0269166	0.0265104	0.00575606	px^-1
roots_per_100px_perimeter	normalised	2.5922	2.3022	1.0111	per 100 px
ring_attached_length_r30_px	network v2	227.562	171.5	116.926	px
ring_attached_length_r60_px	network v2	405.875	309.5	240.962	px
ring_length_w_r30_px	network v2	296.515	255.223	134.101	px
ring_length_w_r60_px	network v2	656.75	556.508	305.847	px
sholl_crossings_r10	network v2	7.1875	5.5	3.8509	count
sholl_crossings_r20	network v2	7.9375	7	4.3277	count
sholl_crossings_r30	network v2	8.3125	6.5	4.2066	count
loop_count	loops, report only	1.5	0	2.3381	count
loop_edge_share	loops, report only	0.0837	0	0.1229	share
ring_width_mean_r30_px	width	7.9068	7.8231	0.7621	px

Per field
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measure	value	units
analysed_area_px2	990645	px^2
total_skeleton_length_px	24632	px
canalicular_length_density_per_px	0.0248646	px^-1
junction_count	430	count
junction_density_per_px2	0.00043406	px^-2
lacuna_count	16	count

lacunae_per_px2	1.615e-05	px^-2
skeleton_component_count	419	count
mean_component_length_px	58.7876	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0280334	px^-1
field_density_without_flagged_per_px	0.0231747	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	1242	px
unattached_component_count	75	count
canalicular_length_density_attached_per_px	0.0248646	px^-1
field_length_density_with_unattached_per_px	0.0261183	px^-1
loop_count	32	count
loop_edge_share	0.0938	share
edge_count	1258	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4996	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19105	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2942	101.888	45.2188	2.2532	0.8961	0.7542	-1.1126	105.425	685.744	False
2	957	54.4409	24.7864	2.1964	0.8903	0.7988	0.0308	119.021	790.769	False
3	1484	55.1992	46.0237	1.1994	0.5521	0.6199	-0.2191	163.815	808.798	False
4	3599	102.364	47.1738	2.1699	0.8875	0.8955	-1.0474	233.848	107.369	False
5	1807	63.8235	38.7415	1.6474	0.7947	0.9372	1.5189	218.143	826.646	False
6	11051	179.07	107.127	1.6716	0.8013	0.7079	0.7029	439.405	253.828	False
7	5312	120.404	59.815	2.0129	0.8679	0.8459	-1.268	460.68	557.419	False
8	1257	48.9366	34.4065	1.4223	0.7111	0.929	1.2464	562.555	134.776	False
9	2123	56.2494	49.4219	1.1381	0.4775	0.9222	0.3101	574.703	493.107	False
10	2990	94.2448	43.1713	2.183	0.8889	0.8771	0.9473	600.532	860.511	False
11	1556	70.3776	32.2729	2.1807	0.8887	0.7988	1.1709	626.462	984.626	False
12	1310	65.9258	27.032	2.4388	0.9121	0.8501	-1.1711	624.014	302.034	False
13	9402	173.447	80.0495	2.1668	0.8871	0.7022	-0.8455	691.761	675.898	False
14	2231	116.941	28.0885	4.1633	0.9707	0.7914	0.9981	758.602	307.963	False
15	6401	142.513	65.0238	2.1917	0.8898	0.8153	-1.2115	832.616	623.668	False
16	3509	107.726	65.7294	1.6389	0.7923	0.6187	-0.7908	851.443	291.978	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	408	877	1151.55	69	16.6891	8.4707
2	4	121	224	134.439	5	26.8877	8.3237
3	7	208	337	255.924	11	23.2658	8.0191

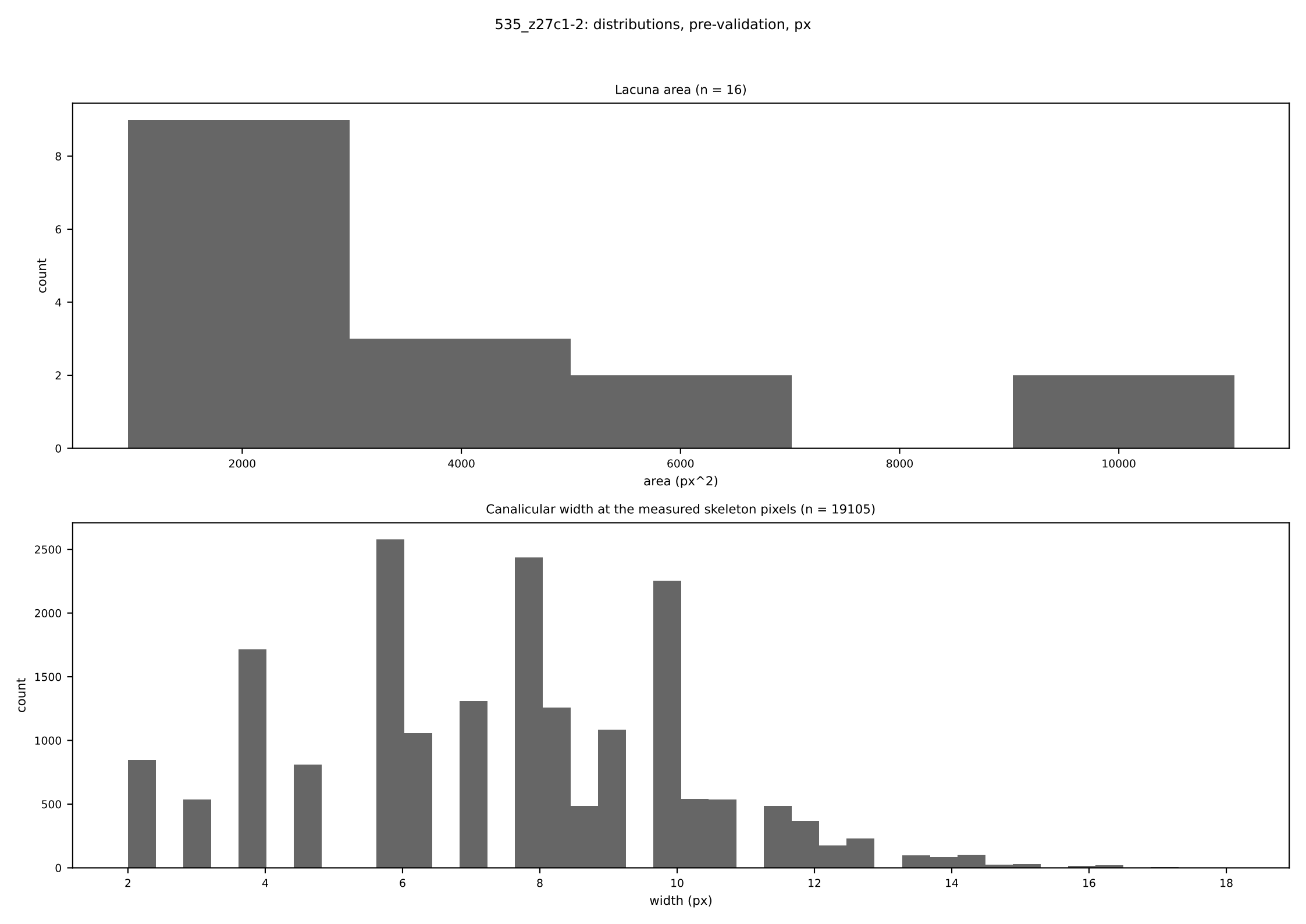
4	6	284	834	1790.48	84	21.3152	7.1459
5	4	140	363	101.154	4	25.2886	7.8006
6	15	445	1043	1159.97	51	22.7446	7.7299
7	18	336	754	935.754	48	19.4949	8.3125
8	7	161	457	525.002	17	30.8825	8.5498
9	5	180	387	295.794	9	32.866	8.4103
10	9	248	520	252.622	11	22.9657	6.7999
11	5	186	402	361.877	17	21.2869	9.9005
12	3	142	269	147.154	4	36.7886	7.8455
13	13	492	1008	2064.58	78	26.469	7.5315
14	4	190	407	331.35	13	25.4885	7.2625
15	8	275	661	761.642	41	18.5766	7.2476
16	10	385	744	1498.33	53	28.2704	7.1584

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	369.203	10657	23260	1	1	0.0382847	0.0377042	2.1668
2	159.095	4461	7968	1	1	0.027124	0.0281125	2.5142
3	254.072	5329	9525	1	1	0.0390317	0.0353806	2.7551
4	281.948	10538	26647	1	1	0.0269501	0.0312981	2.128
5	184.853	6087	14824	1	1	0.0229998	0.0244873	2.1639
6	645.151	19717	40885	1	1	0.0225694	0.0255106	2.325
7	368.818	12630	27408	1	1	0.0266033	0.0275102	4.8805
8	147.054	6972	18793	1	1	0.0230924	0.0243176	4.7602
9	184.267	7991	18400	1	1	0.0225253	0.0210326	2.7135
10	250.392	9454	20849	1	1	0.0262323	0.0249412	3.5944
11	219.35	6971	13409	0.8676	0.7611	0.026682	0.0299799	2.2795
12	195.48	7849	18629	1	1	0.0180915	0.0144399	1.5347
13	634.625	17491	33867	1	1	0.0281288	0.0297635	2.0485
14	284.534	9498	19681	1	1	0.0200042	0.0206798	1.4058
15	465.843	13368	26997	1	1	0.0205715	0.0244842	1.7173
16	402.073	11840	23982	1	1	0.0325169	0.0310233	2.4871

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	382	678	461.818	987.607	10	16	14
2	110	110	140.125	254.137	5	4	2
3	162	228	230.095	376.978	4	7	5
4	224	543	320.108	931.725	6	8	11
5	90	99	161.853	406.576	3	4	6
6	381	718	507.902	1193.57	14	15	15
7	298	631	369.907	837.683	9	10	9
8	157	331	186.823	522.002	5	5	6
9	139	231	199.61	432.433	4	5	8
10	170	262	280.35	591.014	6	7	6
11	173	288	211.581	451.576	5	7	5
12	107	184	157.297	299.279	4	5	4
13	482	888	560.659	1174.86	15	16	15
14	169	254	210.539	456.848	5	5	6
15	255	444	310.693	748.884	13	3	7
16	342	605	434.877	842.826	7	10	14


535_z27c1-2: distributions, pre-validation, px
Lacuna area (n = 16)
count
area (px^2)
0
2
4
6
8
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 19105)
count
width (px)
0
500
1000
1500
2000
2500
2
4
6
8
10
12
14
16
18